



Stomach II — Polyps & Gastric Cancer

Pathology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. Which gastric polyp is defined by **dysplasia**, and what does that mean for its risk?
2. Which Lauren type has **no precursor lesion and no risk factors**?
3. Name **three** metastatic eponyms of gastric carcinoma and their sites.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Gastric polyps

A polyp is **any nodule or mass projecting above the surrounding mucosa**. Gastric polyps are **uncommon compared with colonic polyps**.

Type	Association	Malignant potential
Hyperplastic / inflammatory	The commonest — arise in chronic gastritis	Low
Fundic gland	FAP and chronic PPI use	Low
Adenomatous	Defined by the presence of dysplasia	Definite risk of harbouring adenocarcinoma

Dysplasia is the dividing line. Adenomatous polyps are defined by it — which is exactly why they carry real cancer risk while the other two do not.

2 Gastric cancer — the frame

Commonest	Adenocarcinoma — over 90% of all gastric cancers.
Lymphoma	The stomach is the commonest site for extranodal NHL .
Location	Most involve the gastric antrum ; the lesser curvature more often than the greater.
Geography	Incidence varies markedly by region.
Falling rates	Dropped by more than 85% during the 20th century — attributed to less dietary N-nitroso compounds and benzo[a]pyrene as salting and smoking of food declined, plus mass endoscopic screening in high-incidence regions detecting early cancers.

Risk factors

The block's recurring pattern, in the stomach:

H. pylori → chronic gastritis → **gastric atrophy** → **intestinal metaplasia** → **epithelial dysplasia** → **carcinoma**.

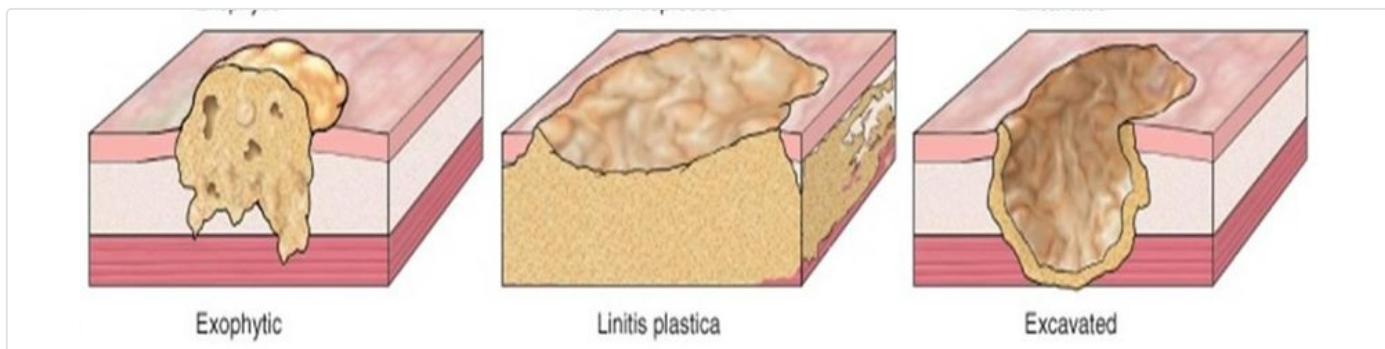
Same shape as GERD → Barrett → adenocarcinoma and adenoma → colorectal cancer.

Infection	H. pylori , via the sequence above
Predisposing disease	Atrophic gastritis , autoimmune gastritis, pernicious anaemia · gastric adenomatous polyps · subtotal gastrectomy · gastric ulcer
Lifestyle / social	Smoking · lower socioeconomic groups
Chemical	Nitrosamine
Diet — increases risk	Large amounts of starch ; smoked or cured fish and meat; pickled vegetables
Diet — decreases risk	Whole milk ; fresh vegetables rich in vitamin C and other antioxidants

Presentation is non-specific and late: **abdominal discomfort**, **early satiety**, **nausea and vomiting**, **GI haemorrhage**, **iron deficiency anaemia**.

3 Gross morphology

Pattern	Description
Polypoidal (exophytic)	Projects into the lumen
Infiltrative	Linitis plastica — diffuse thickening of the wall
Ulcerative (excavated)	Malignant ulcer — irregular, heaped-up borders and a dirty necrotic base



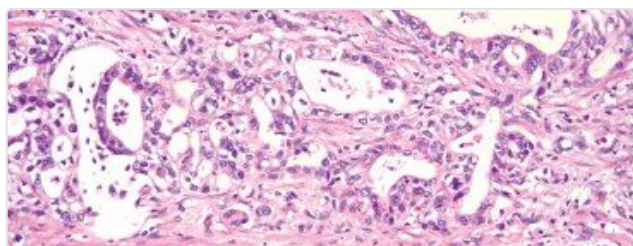
Exophytic · linitis plastica · excavated

Benign vs malignant ulcer: a benign peptic ulcer has **regular margins, radiating folds and a clean base**; a malignant one has **heaped-up irregular borders and a necrotic base**. This comparison is a classic exam item.

4 Lauren classification — intestinal vs diffuse

Feature	Intestinal type	Diffuse type
Histology	Glandular structures , similar to esophageal and colonic adenocarcinoma	Signet ring cell morphology
Growth	Exophytic mass or ulcerated tumour	Infiltrative
Precursor lesion	Usually arises from a precursor lesion	No precursor lesion
Risk factors	Has recognised risk factors	No known risk factors
Stromal reaction	—	Desmoplastic reaction stiffens the wall → rugal flattening → "leather bottle" = linitis plastica

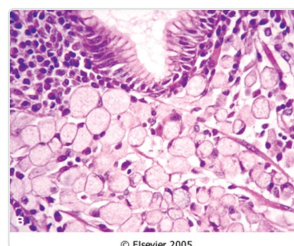
Why this matters clinically: the intestinal type has a precursor and risk factors, so it is the one screening and H. pylori eradication can prevent. The diffuse type has neither — it arises without warning, which is part of why it behaves badly.



Intestinal type — malignant cells forming glands, invading muscle



Linitis plastica — diffusely thickened, rigid wall



Diffuse type — signet ring cells

5 Metastases — the eponyms

Often already present at diagnosis.

Site	Eponym
Supraclavicular lymph node	Virchow node / Troisier sign
Periumbilical region	Sister Mary Joseph nodule
Left axillary lymph node	Irish node
Ovary	Krukenberg tumour
Pouch of Douglas	Blumer shelf



Virchow node — left supraclavicular



Sister Mary Joseph nodule — periumbilical



Krukenberg tumour — bilateral ovarian deposits

Five eponyms, five locations. They are pure recall marks — cover the right column and rebuild the list until it is automatic.

P Must-remember pearls

- **Adenocarcinoma is over 90% of gastric cancer**; the stomach is also the **commonest site of extranodal NHL**.
- **H. pylori → gastritis → atrophy → intestinal metaplasia → dysplasia → carcinoma**. Learn the chain, not the endpoints.
- **Intestinal type has a precursor and risk factors; diffuse type has neither** and shows signet ring cells with linitis plastica.
- **Adenomatous polyps are defined by dysplasia** — the reason they carry real cancer risk.
- **Malignant ulcer = heaped-up irregular borders, necrotic base**. Benign = regular margins, radiating folds, clean base.
- **Krukenberg = ovary, Virchow = supraclavicular, Sister Mary Joseph = umbilicus, Irish = left axilla, Blumer = pouch of Douglas**.

Q Test yourself

- 1. A gastrectomy specimen shows a diffusely thickened, rigid stomach wall with flattened rugae. Histology shows signet ring cells. Which statement is correct?**

A. It typically arises from a recognised precursor lesion	B. It has no known precursor lesion or risk factors
C. It forms glands resembling colonic adenocarcinoma	D. It usually presents as an exophytic mass
- 2. A woman with gastric adenocarcinoma is found to have bilateral ovarian masses. What is this called?**

A. Blumer shelf	B. Irish node
C. Krukenberg tumour	D. Sister Mary Joseph nodule
- 3. Which gastric polyp carries a definite risk of harbouring adenocarcinoma?**

A. Hyperplastic polyp	B. Fundic gland polyp
C. Adenomatous polyp	D. Inflammatory polyp
- 4. Gastric adenocarcinoma incidence fell by more than 85% in the twentieth century. Which factor is the most likely contributor?**

A. Increased use of PPIs	B. Reduced dietary N-nitroso compounds from less salting and smoking of food
C. Wider use of subtotal gastrectomy	D. Rising consumption of pickled vegetables
- 5. Endoscopy shows a gastric ulcer with heaped-up irregular margins and a dirty necrotic base. Compared with a benign peptic ulcer, which feature is most suggestive of malignancy?**

A. Solitary lesion	B. Location on the lesser curvature
C. Heaped-up irregular borders	D. Size under 2 cm

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 • B** — Linitis plastica with signet ring cells is the diffuse type, which characteristically has no precursor lesion and no known risk factors — the opposite of the intestinal type.
- 2 • C** — Krukenberg tumour is metastatic gastric carcinoma in the ovary, classically bilateral.
- 3 • C** — Adenomatous polyps are defined by dysplasia and carry a definite risk of harbouring adenocarcinoma; hyperplastic (the commonest) and fundic gland polyps do not.
- 4 • B** — Reduced salting and smoking of food lowered intake of N-nitroso compounds and benzo[a]pyrene; mass endoscopic screening also detected early cancers.
- 5 • C** — Benign peptic ulcers have regular margins with radiating folds and a clean base; heaped-up irregular borders with a necrotic base indicate malignancy.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pathology GIT Block · MEDucation by Mattin Majid · summarised from SGL1, Dr Jawher Tahir Omer, Hawler Medical University
Study alongside the lecture, not instead of it.